2 - Probability

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Introduction

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Scan the QR code to submit your email

MH1820 Tutorial

https://forms.cloud.microsoft/r/Y5jgUkwqmB

Additional Resources (Dr Chunfei)

tinyurl.com/1820dropbox

- Additional Exercises
- Finals/Midterm prep
- Recap slides
- Tutorial Solutions (Handwritten)

Note: The Youtube link in the folder have been deprecated; use this instead.

https://www.youtube.com/playlist?list=PLTFRGgGewwzIrMiLFtXVHpD8PvYPk-qGz
(Only from Tut 5 onwards)

Recap: Conditional Probability & Indept

Let $A, B \subseteq \Omega$ be events with $\mathbb{P}(A) > 0$. The **conditional probability** of B given A is

$$\mathbb{P}(B|A) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(A)}.$$

Note that $\mathbb{P}(A \cap B) = \mathbb{P}(B|A)\mathbb{P}(A)$. This formula often can be used to compute $\mathbb{P}(A \cap B)$.

Formally, events A and B are **independent** if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

Problem 1 (sample spaces)

For each of the following statistical experiments, write down a suitable sample space.

(i) The number of passengers on an MRT train is counted.

(ii) The height of a person is measured.

(iii) 1000 persons are selected randomly and they are asked if they have any travel plans for this year.

(iv) The maximum force (in N=Newton) is determined that a certain steel beam can withstand without breaking.

(v) At a random time, the number of seconds past the current minute is determined (for instance, if the current time is 12:13:33, the outcome is 33).

(vi) A dice is rolled 5 times.

$\boxed{\textbf{Solution}}$ Sample spaces are not uniquely determined, so the solutions given here are not the only valid ones. Typically, it is best to keep the sample space as simple as possible while making sure that it represents all possible outcomes of the experiment.

(i) Here the simplest choice for the sample space is $\Omega = \mathbb{Z}_0^+ = \{0, 1, 2, \ldots\}$ (set of nonnegative integers). An outcome $x \in \Omega$ means that there are x passengers on the train.

Since usually there are less than 2000 passengers on an MRT train, we could also choose $\{0, 1, \ldots, 2000\}$ as the sample space, but an upper bound like 2000 is not useful for calculating probabilities. When large numbers of possible outcomes are involved, we typically only use *approximations* of probabilities (not exact probabilities) and for these approximations we do not need an upper bound.

(ii) The height of a person can be represented by a nonnegative real number (the height in centimeters), so we can take $\Omega = \mathbb{R}_0^+$ (set of nonnegative real numbers) as the sample space. Again, we could introduce an upper bound (say $\Omega = [0, 300]$, the interval from 0 to 300), but this is not necessary.

(iii) Let $P_1, \ldots, P_{1000}$ be the selected persons and set $x_i = 0$ if P_i answers no and $x_i = 1$ if P_i answers yes. Given this setting, Ω can be taken as the set of all 1000-tuples $(x_1, \ldots, x_{1000})$ with $x_i \in \{0, 1\}$. Note that this sample space is huge ($|\Omega| = 2^{1000}$).

(iv) Given no further information, the best choice for the sample space is $\Omega = \mathbb{R}_0^+$.

(v) The possible outcomes are $0, 1, \ldots, 59$ and we can take $\Omega = \{0, 1, \ldots, 59\}$. If the measurement was more precise, say up to milliseconds, then $\Omega = [0, 60]$ (the interval of real numbers from 0 to 60) would be more appropriate.

(vi) Let $x_i \in \{1, \ldots, 6\}$ be the outcome of the ith roll. In this setting, the sample space Ω is the set of all 5-tuples $(x_1, \ldots, x_5)$ with $x_i \in \{1, \ldots, 6\}$. Note that $|\Omega| = 6^5$.

Problem 2 (independent events)

Two integers are drawn independently at random from $1, 2, \ldots, 100$ (repetition is allowed, for instance, both numbers could be 1). Consider the following events.

E_1: both numbers are even
E_2: the first number is ≤ 50
E_3: the second number is prime

For any pair of events above, decide whether the two events are independent.

Clue: There are **25** prime numbers from 1-100

Problem 3 (conditional probability)

A researcher finds that, of 982 men who died in 2002, 221 died from some heart disease. Also, of the 982 men, 334 had at least one parent who had some heart disease. Of the latter 334 men, 111 dies from some heart disease. A man is selected from the group of 982. Given that neither of his parents had some heart disease, find the conditional probability that this man died of some heart disease.

Problem 4 (conditional probability)

(a) A dice is rolled 4 times.

(i) What is the probability that **at most** one of the rolls is a 6?

(ii) What is the probability that **at least** two of the rolls are a 6?

(iii) What is the probability that the total of the rolls is **at least** 22?

(iv) What is the probability that the total of the rolls is **at least** 22 under the condition that at least two of the rolls are a 6?

(b) A family has two children. Assume that the probability for a child to be a girl is $\frac{1}{2}$, and the probability for a child to be a boy is $\frac{1}{2}$.

(i) What is the probability that both children are girls under the condition that the first child is a girl?

(ii) What is the probability that both children are girls under the condition at least one of the children is a girl?

Recap: Bayes' Theorem

Theorem 8 (Bayes' Theorem)

Let A, B be events with $\mathbb{P}(A), \mathbb{P}(B) > 0$. Then

$$\mathbb{P}(A|B) = \frac{\mathbb{P}(B|A)\mathbb{P}(A)}{\mathbb{P}(B)}.$$

Problem 5 (law of total probability and Bayes' theorem)

A ball is drawn from one of 3 boxes. The boxes contain the following number of balls of colors blue (B), red (R), and yellow (Y).

	B	R	Y
Box 1	1	4	5
Box 2	6	2	2
Box 3	3	3	4

The following procedure is used to draw the ball.

- One of the boxes is chosen at random: Box 1 is chosen with probability 0.2, Box 2 is chosen with probability 0.5, and Box 3 is chosen with probability 0.3.
- A ball is drawn from the chosen box (each ball in the box is chosen with the same probability).

(i) What is the probability that a blue ball is drawn?

(ii) What is the probability that a yellow ball is drawn?

(iii) If a blue ball is drawn, what is the probability that it was drawn from Box 1?

Bonus Qn: Defective Radios

(5) Two local factories, A and B, produce radios. Each radio produced at factory A is defective with probability 0.05, whereas each one produced at factory B is defective with probability 0.01. Suppose you purchase two radios that were produced at the same factory, which is equally likely to have been either factory A or factory B. If the first radio that you check is defective, what is the conditional probability that the other one is also defective?